

The Emergence of Relational Time (Page-Wootters Mechanism)

1. Global state

no external time

$$|\Psi\rangle$$
$$\hat{H}|\Psi\rangle = 0 \text{ (static Wheeler-DeWitt block)}$$

The field state carries correlations,
not a time parameter.

2. Decomposition

clock-system split

$$|\Psi\rangle = \sum_n |n\rangle_c \otimes |\psi(n)\rangle_s$$



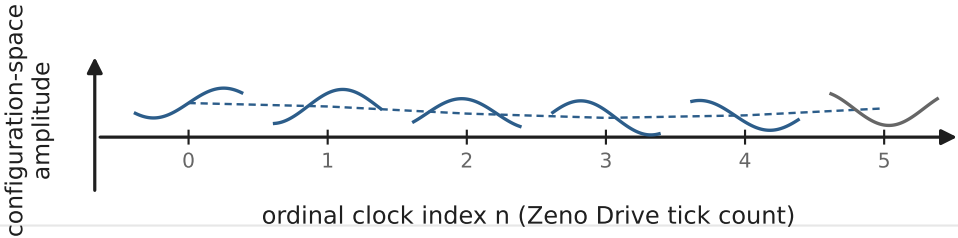
3. Conditional readout

agent view



4. Reconstructed sequence

ordinal dynamics



Dynamics are recovered as conditional correlations $\{|\psi(n)\rangle_s\}$ inside the frozen global state.